



IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

Core Java

Virtual Internship

AICTE – UpSkill Campus & UniConverge Technologies Pvt. Ltd.

Anusha Kumari

Roll No. 25SCS1003002909 | B.Tech CSE AI-ML

July – August 2026



**Anusha
Kumari**

B.Tech – Computer Science & Engineering

ABOUT THE STUDENT

Candidate Profile



Roll Number

25SCS1003002909



Institute

IILM University, Greater Noida, U.P.



Programme

B.Tech CSE AI-ML, Batch 2025–2029



Internship Domain

Core Java



Duration

4-6Weeks | July – August 2026

ABOUT THE PROGRAM

AICTE , UpSkill Campus & UniConverge Technologies

UpSkill Campus, along with its industry partner UniConverge Technologies Pvt. Ltd. (UCT), provides industry-aligned technical internships in collaboration with the All India Council for Technical Education (AICTE). The Core Java Virtual Internship track delivers a hands-on, project-driven curriculum designed to bridge the gap between academic theory and real-world software engineering through milestone-based development of console applications.



Industry-Aligned Track

Practical software development curriculum powered by UniConverge Technologies.



AICTE Collaboration

Conducted under the patronage of the AICTE National Internship initiative.



Hands-On Core Java

Milestone-based project development focusing on OOP, Streams, and File I/O.

Bridging Theory and Applied Java Development

Although many engineering students learn the theoretical foundations of Java and OOP in coursework, very few get structured, hands-on exposure to building persistent, modular multi-tier applications without external databases.

Key Gaps:

Students learn Java syntax but lack experience with real-world application architecture

Limited exposure to OOP encapsulation, separation of concerns, and layered design patterns

No hands-on practice with data persistence using file-based storage mechanisms

Gap between console-based learning and enterprise-grade application patterns

Solution: Build a production-grade Expense Tracker with multi-tier architecture, CSV persistence.

What the Internship Set Out to Achieve

Six Core Learning Outcomes

OOP Encapsulation Implement domain models with proper access modifiers and data hiding.

Full CRUD Lifecycle Complete Create, Read, Update, Delete operations for expenses.

Java 8 Streams Advanced data analytics using functional programming and lambdas.

Budget Alerts Real-time monthly budget tracking with warning notifications.

CSV Persistence Reliable flat-file data storage without external database dependencies.

Resilient UI User-friendly console interface with defensive input validation.

The Toolkit Behind the Internship

Core Java 17+: Latest language features and performance optimizations

OOP :Inheritance, polymorphism, encapsulation, abstraction

Collections :ArrayList for dynamic expense storage

Streams & Lambdas: Functional programming for data filtering

Date/Time API :YearMonth for monthly budget cycles

File I/O: BufferedReader/Writer for CSV operations

Exception Handling: Try-catch and custom exceptions

Dev Tools: VS Code, Git/GitHub, run.bat automation

3-Tier Layered Architecture



The 4-Week Learning Journey

1

WEEK

Foundation & Setup

OOP fundamentals, Git setup, project scaffolding, and version control workflow

2

WEEK

Core Development

Domain model design, full CRUD operations, and CSV persistence implementation

3

WEEK

Advanced Features

Java 8 Streams for analytics, YearMonth-based filtering, and budget alert logic

4

WEEK

Polish & Release

Console UI refinement, budget warning banners, comprehensive testing, and release

Skills & Capabilities Gained



Multi-Tier Architecture Design: Separation of concerns and layered application structure



Stream Analytics: Functional programming for complex data filtering and aggregation



Flat-File Storage: CSV persistence without relational databases



Defensive Programming: Comprehensive exception handling and error recovery



Input Validation: Fail-safe user input processing and sanitization



Professional Code Quality: Clean code, documentation, and best practices



Version Control Mastery: Git workflows and collaborative development



Enterprise Patterns: Real-world application design and deployment readiness

CERTIFICATION

Successfully Completed



Completion Status

Successfully Completed



Certificate ID

USC949986TIA



GitHub Repository Link

<https://github.com/anushasingh17/Expense-Tracker-Application.git>



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Thank You

Anusha Kumari Virtual Internship | AICTE – UpSkill Campus

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